

FORMAL LANGUAGES, AUTOMATA AND COMPUTATION

PUMPING LEMMA

PROPERTIES OF CFLs

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SUMMARY

- Context-free Languages and Context-free Grammars
- Pushdown Automata
- PDAs accept all languages CFGs generate.
- CFGs generate all languages that PDAs accept.
- There are languages which are NOT context free.

PUMPING LEMMA FOR CFLs

LEMMA

If L is a CFL, then there is a number p (the pumping length) such that if s is any string in L of length at least p , then s can be divided into 5 pieces $s = uvxyz$ satisfying the conditions:

- 1 $|vy| > 0$
- 2 $|vxy| \leq p$
- 3 for each $i \geq 0$, $uv^i xy^i z \in L$

- The pumping length is determined by the number of variables the grammar for L has.

APPLICATION OF THE PUMPING LEMMA

- Just as for regular languages we employ the pumping lemma in a two-player game setting.
- If a language violates the CFL pumping lemma, then it can not be a CFL.
- Two Player Proof Strategy:
 - Opponent picks p , the pumping length
 - Given p , we pick s in L such that $|s| \geq p$. We are free to choose s as we please, as long as those conditions are satisfied.
 - Opponent picks $s = uvxyz$ - the decomposition subject to $|vxy| \leq p$ and $|vy| \geq 1$.
 - We try to pick an i such that $uv^ixy^iz \notin L$
 - If for all possible decompositions the opponent can pick, we can find an i , then L is not context-free.

USING PUMPING LEMMA – EXAMPLE-1

- Consider the language $L = \{a^n b^n c^n \mid n \geq 0\}$
- Opponent picks p .
- We pick $s = a^p b^p c^p$. Clearly $|s| \geq p$.
- Opponent may pick the string partitioning in a number of ways.
- Let's look at each of these possibilities:

USING PUMPING LEMMA—EXAMPLE 1

- Cases 1,2 and 3: vxy contains symbols of only one kind

① Only a 's: $\underbrace{a \cdots a}_u \underbrace{a \cdots a}_{vxy} \underbrace{a \cdots ab \cdots bc \cdots c}_z$

② Only b 's: $\underbrace{a \cdots ab}_u \underbrace{b \cdots b}_{vxy} \underbrace{b \cdots bc \cdots c}_z$

③ Only c 's: $\underbrace{a \cdots ab \cdots bc}_u \underbrace{c \cdots c}_{vxy} \underbrace{c \cdots c}_z$

- Pumping v and y will introduce more symbols of one type into the string.
- The resulting strings will not be in the language.

USING PUMPING LEMMA—EXAMPLE 1

- Cases 4 and 5: vxy contains two symbols – crosses symbol boundaries.

① Only a 's and b 's: $\underbrace{a \cdots a}_u \underbrace{a \cdots ab \cdots b}_{vxy} \underbrace{b \cdots bc \cdots c}_z$

② Only b 's and c s: $\underbrace{a \cdots ab \cdots b}_u \underbrace{b \cdots c}_{vxy} \underbrace{c \cdots c}_z$

- Note that vxy has length at most p so can not have 3 different symbols.
- Pumping v and y will both upset the symbol counts and the symbol patterns.
- The resulting strings will not be in the language.

FINAL

USING PUMPING LEMMA—EXAMPLE 2

- Consider the language $L = \{a^n \mid n \text{ is prime}\}$
- Opponent picks p .
- We pick $s = a^p$. Clearly $|s| \geq p$.
- Opponent may pick any partitioning $s = uvxyz$.
 - Let $m = |uxz|$ for the partitioning selected, that is, **the length of everything else but v and y .**
 - Any pumped string $uv^i xy^i z$ will have length $m + i(p - m)$.
 - We choose $i = p + 1$.
 - The pumped string has length $m + (p + 1)(p - m)$. But:

$$\begin{aligned}m + (p + 1)(p - m) &= m + p^2 - pm + p - m \\&= p^2 + p - pm \\&= p(p - m + 1)\end{aligned}$$

which is **not prime** since both p and $p - m + 1$ are greater than 1. (Note $0 \leq m \leq p - 1$)

CLOSURE PROPERTIES OF CONTEXT-FREE LANGUAGES

- Context-free languages are closed under
 - Union
 - Concatenation
 - Star Closure
 - Intersection with a regular language
- We will provide very informal arguments for these.

CLOSURE PROPERTIES OF CFLS-UNION

- Let G_1 and G_2 be the grammars with start variables S_1 and S_2 , variables V_1 and V_2 , and rules R_1 and R_2 .
- Rename the variables in V_2 if they are also used in V_1
- The grammar G for $L = L(G_1) \cup L(G_2)$ has
 - $V = V_1 \cup V_2 \cup \{S\}$ (S is the new start symbol $S \notin V_1$ and $S \notin V_2$)
 - $R = R_1 \cup R_2 \cup \{S \rightarrow S_1 \mid S_2\}$

CLOSURE PROPERTIES OF CFLS – CONCATENATION

- Let G_1 and G_2 be the grammars with start variables S_1 and S_2 , variables V_1 and V_2 , and rules R_1 and R_2 .
- Rename the variables in V_2 if they are also used in V_1
- The grammar G for $L = \{wv \mid w \in L(G_1), v \in L(G_2)\}$ has
 - $V = V_1 \cup V_2 \cup \{S\}$ (S is the new start symbol $S \notin V_1$ and $S \notin V_2$)
 - $R = R_1 \cup R_2 \cup \{S \rightarrow S_1 S_2\}$

CLOSURE PROPERTIES OF CFLS – STAR CLOSURE

- Let G_1 be the grammar with start variable S_1 , variables V_1 , rules R_1 .
- The grammar G for $L = \{w \mid w \in L(G_1)^*\}$ has
 - $V = V_1 \cup \{S\}$ (S is the new start symbol $S \notin V_1$).
 - $R = R_1 \cup \{S \rightarrow S_1 S \mid \epsilon\}$

CLOSURE PROPERTIES OF CFLS – INTERSECTION WITH A REGULAR LANGUAGE

- Let P be the PDA for the CFL L_{cfl} and M be the DFA for the regular language $L_{regular}$
- We have a procedure for building the **cross-product PDA** from P and M .
 - Very similar to the cross-product construction for DFAs.
 - Details are not terribly interesting. (Perhaps later.)

CLOSURE PROPERTIES OF CFLs

- CFLs are NOT closed under intersection.
 - $L_1 = \{a^n b^n c^m \mid n, m \geq 0\}$ is a CFL.
 - $L_2 = \{a^m b^n c^n \mid n, m \geq 0\}$ is a CFL.
 - $L = L_1 \cap L_2 = \{a^n b^n c^n \mid n \geq 0\}$ is NOT a CFL.
 - CFLs are not closed under complementation.
 - $L = \{ww \mid w \in \Sigma^*\}$ is NOT a CFL (Prove it using pumping lemma!)
 - $\bar{L}$ is actually a CFL and $L = L_1 \cup L_2$
 - $\bar{L}$ has all strings of odd length (L_1)
 - $\bar{L}$ has all strings where at least one pair of symbols $n/2$ apart are different (n length of the string!) (L_2)
- | | |
|--|----------------------------|
| $S \rightarrow aA \mid bA \mid a \mid b$ | $S \rightarrow AB \mid BA$ |
| $A \rightarrow aS \mid bS$ | $A \rightarrow ZAZ \mid a$ |
| generates L_1 | $B \rightarrow ZBZ \mid b$ |
| | $Z \rightarrow a \mid b$ |
| | generates L_2 |

CFL CLOSURE PROPERTIES IN ACTION

- Is $L = \{a^n b^n \mid n \geq 0, n \neq 100\}$ a CFL?
 - $L = \underbrace{\{a^n b^n \mid n \geq 0\}}_{CFL} \cap \underbrace{(L(a^* b^*) - \{a^{100} b^{100}\})}_{RL}$
 - The intersection of a CFL and a RL is a CFL!
- Is $L = \{w \mid w \in \{a, b, c\}^* \text{ and } n_a(w) = n_b(w) = n_c(w)\}$ a CFL?
 - $\underbrace{L}_{CFL?} \cap \underbrace{L(a^* b^* c^*)}_{RL} = \underbrace{\{a^n b^n c^n \mid n \geq 0\}}_{\text{Not CFL}}$
 - Thus L is NOT a CFL.

MOVING BEYOND THE MILKY WAY

WHAT OTHER KINDS OF LANGUAGES ARE OUT THERE?

How can we characterize these languages outside the boundary of CFLs?

